KLEO-1: UCF's First Student Satellite

Building an Accessible, Student-Led Pathway from Tethered Balloons to Low Earth Orbit

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Abstract

- **Knights Satellite Club (KSC)** prepares UCF students for **orbital missions** through tethered and high-altitude balloons, ground stations and hands-on engineering.
- Its first CubeSat, **KLEO-1**, uses the **PROVES** platform to **receive telemetry from low Earth orbit**.
- The program builds student expertise and **repeatable processes** for future missions.

Background

- KSC is a **registered student organization** at UCF, founded in **2020** with the goal of launching a **student-built satellite**.
- **Lower launch costs** and **commercial off-the-shelf (COTS)** components make student orbital missions increasingly feasible.
- Yet university CubeSat programs often stall from **long development cycles**, funding gaps and **lost institutional knowledge**.

4 of ~35 university CubeSat teams tracked since 2019 had launched, per the PROVES developers [1].

- KSC answers with **short, low-cost missions** that build skills and **member retention** before committing to orbit.

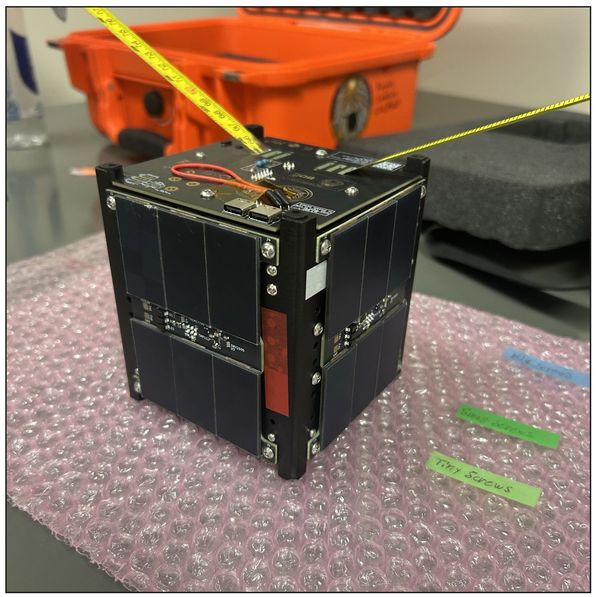


Figure 1. PROVES V3 engineering unit used for KLEO-1.

The KSC Project Pipeline

- **Skill progression** is the pillar of the pipeline: each stage adds engineering skills that map to **university coursework**, so members always have a clear next goal.

STAGE 1	ALTITUDE		STAGE 2	ALTITUDE		STAGE 3	ALTITUDE
TSat	**~150 m**	×200 →	**KAOS**	**~30 km**	×13 →	**KLEO-1**	**~400 km**
Moored balloon on a tether			High-altitude balloon · stratosphere			PROVES CubeSat · low Earth orbit	
Integration, embedded systems, communications			*Environmental testing, flight ops, recovery*			*Spacecraft integration, qualification, operations*	

Ground stations support every stage: tracking, telemetry reception and mission operations

AUG 2025	SPRING 2026	FALL 2026	JAN 2027 (TARGET)	END 2027 (TARGET)
KAOS-1 flown	TSat-2A and TSat-2B flown	KAOS-2 flight; KLEO-1 EDU testing	TSat-3A and TSat-3B launch	KLEO-1 launch-ready

Figure 2. KSC's three-stage mission pipeline, from a ~150 m tether to ~400 km orbit, and program timeline. Typical altitudes shown.

Stage 1 · Tethered (TSat)

- The **entry point for new members**: payloads fly on a moored balloon and go from **design to launch in one or two semesters**.
- **TSat-2A** tested a **2U housing**, cameras and radio-commanded **servo separation**, reaching **~450 ft** and recovering image and altitude data.
- **TSat-2B** streamed **live video** (OpenHD) and **live telemetry** (ESP32 + RFM69) for its entire flight.
- **TSat-3A** adds **reaction-wheel stabilization**; **TSat-3B** adds automated **parafoil return to launch site**.

Stage 2 · High Altitude (KAOS)

- **KAOS-1** launched **Aug 30, 2025** from the Econ River Wilderness Area.
- It logged **temperature, pressure, GPS, orientation, CO_2** and **humidity**, and demonstrated **communication**, **power** and **recovery** systems built to mirror orbital constraints.
- **KAOS-2** adds **APRS** communications, a redesigned structure and **thermal testing**.

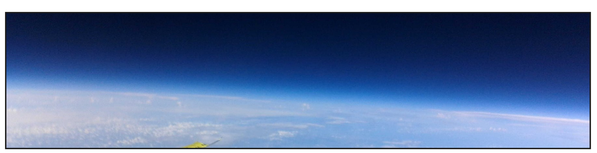


Figure 3. Earth's curvature from KAOS-1, Aug 30, 2025.

Stage 3 · Orbit (KLEO-1)

- Built on the established **PROVES** architecture so students can focus on **integration, testing and operations**.
- First in a planned **series of CubeSats** of increasing complexity and scientific value.
- **Primary objective:** receive telemetry from **low Earth orbit**.

Platform	PROVES Kit (Bronco Space)
Hardware	V3 EDU unit; two V3.1 kits ordered
Attitude	Magnetorquers, magnetometer
Software	F Prime (PROVES fork)
Payload	In selection; interfaces in work
Launch	Commercial rideshare (TBD)



Figure 4. KSC members integrating the PROVES unit.

Ground Stations

- Support **every stage** of the pipeline, from balloon tracking to orbital operations.
- The **Physics Ground Station**, run with the UCF Department of Physics, sits on the Physical Sciences Building roof (call sign **KQ4OHW**).
- **2 m** and **70 cm** yagis on a computer-controlled **rotator** track satellites with **GPredict**; a **TinyGS** node is prioritized for upcoming PROVES launches.

Lessons Learned (TSat-2A)

- Test deployment **end to end** with final flight hardware.
- **Freeze the design early** to leave time for late fixes.
- Verify critical actions **physically or with a sensor**, not just a software acknowledgement.

Future Work

- Fly **KAOS-2** and launch **TSat-3A** and **TSat-3B** together (target Jan 2027).
- Select the KLEO-1 **payload** and complete **flight software** bring-up.
- Have KLEO-1 **launch-ready by end of 2027**.

Conclusions

- A **staged pipeline** takes students from no satellite experience to **orbital mission work**.
- **Open designs** and **public documentation** lower the barrier for future teams at UCF and beyond.
- KSC offers a practical model for expanding access to satellite engineering within the **SpaceU ecosystem**.

References

[1] Pham et al., "The Any% Method: Improving Space Access Through Improved Design, Build, and Test Methodologies," 38th Annual Small Satellite Conf., SSC24-WVII-09, 2024.

[2] Bronco Space, PROVES CubeSat Kit, docs.proveskit.space.

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